

Notice to Prospective Offerors

This document provides the required instructions, testing procedures, and evaluation criteria for vendors submitting PIT tags for performance testing under this solicitation. All offerors must review and follow the procedures outlined below to ensure consistent, fair, and accurate evaluation of tag performance.

General Instructions

- Vendors must perform **all** tests described in this protocol.
- Vendors must supply **all necessary testing equipment**.
- All testing will be conducted in the presence of designated Government personnel, which may include:
 - Contracting Officer Representative (COR)
 - Contracting Officer (CO)
 - Contract Specialist
 - Technical Proposal Evaluation Committee members
- All tests will be performed using a **Biomark IS1001 reader**, the standard reader used by most Department of the Interior agencies for field PIT tag detection.
- For each test, tags will be **randomly selected** from a batch of 1,000 tags provided by the vendor.
- The required sample size for each test is specified within each section.
- Testing will stop immediately if a tag set fails any pass/fail threshold stated in this protocol.

Offerors are responsible for understanding the criteria, ensuring accurate measurements, and providing complete documentation of results.

PIT Tag Testing Protocol

1. Physical Specifications (Length, Weight, and Diameter)

All tags will undergo physical inspection for structural integrity, including checks for broken housings, missing components, or abnormal shapes.

Measurement Requirements

- **Sample size:** 100 tags
- **Length and diameter:**
 - Measured to the nearest 0.01 mm using micrometers/calipers
 - Tags must pass freely through a standard 12-gauge veterinary hypodermic needle
 - **Failing criteria:**
 - Average length > 12.7 mm (or > 9.3 mm for shorter standard tags)
 - Average diameter > 2.14 mm
- **Weight:**
 - Measured to the nearest 0.0001 g using an electronic analytical balance
 - **Failing criteria:**
 - Average weight > 0.115 g when measured in air

Evaluation Notes

- Tags meeting the preferred length (≤ 12.7 mm) and weight (≤ 0.115 g) will receive favorable consideration during proposal evaluation.
 - If tags fail length, diameter, or weight requirements, testing will stop.
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2. PIT Tag Electrical Performance Test

This test evaluates electrical characteristics affecting activation and transmission performance.

Tags are tested using a sine wave signal to measure:

- **Turn-On Voltage (mV):** Minimum energy required for activation
- **Resonant Frequency (kHz):** Peak transmission frequency
- **Bandwidth (kHz):** Transmission signal width at -3 dB
- **Modulation (%):** Effectiveness of transmitting unique identifier

Sample Size

- Minimum of 200 PIT tags.

Passing Requirements

Testing will stop if fewer than **98%** of tags meet all of the following:

- Resonant Frequency: **132–136.5 kHz**
 - Turn-On Voltage: **≤ 700 mV**
 - Modulation: **> 75%**
 - Bandwidth: **< 9 kHz**
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3. Interfering Distance Test

This test measures detection efficiency when tags are in close proximity to one another.

Method

- **Sample size:** Five sets of two randomly selected tags
- Tags are run through a **12-inch antenna** at a high rate of speed (i.e. 13 ft/sec)
- Tag spacing distances: **6 inches** and **3 inches**
- Each pair is run through **two antennas, 100 times each**, for a total of 2,000 detection opportunities per tag type.

Passing Requirement

- Efficiency must be **≥ 95%** for both spacing distances.
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4. Minimum Read Distance Test

This test determines the distance at which a PIT tag can be successfully detected.

Method

- Conducted using a **4' x 4' antenna**
- Orientation angles tested: **0°, 45°, and 66°** (66° is the most difficult)
- Noise conditions tested: **0% and 10%**
- All testing occurs in a fully shielded, noise-free environment

Passing Requirement

- Tags must have a minimum read distance of **8” for 0° orientation** and **6” for 45° orientation** at a **10% noise level**
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5. Read Speed Test

This test evaluates tag readability at high movement speeds.

Method

- **Sample size:** 10 randomly selected tags
- Tags passed through a **12-inch shielded antenna** at high rate of speed (~25 ft/sec)
- Minimum of **100 passes through** per tag
- Testing includes:
 - Tag orientations of **0°** and **45°**
 - **25% noise** to differentiate tag performance under adverse conditions

Passing Requirement

- Tags must maintain **≥ 95% efficiency**.
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Submission Requirements

Vendors must submit:

- A complete test report documenting methods, equipment, data, and results
- Signed certification that all testing was performed according to this protocol
- Identification of any deviations or anomalies observed during testing